

Futsal Arena: Futsal Management System

Minor Project Defense | Backend-Driven Web Application

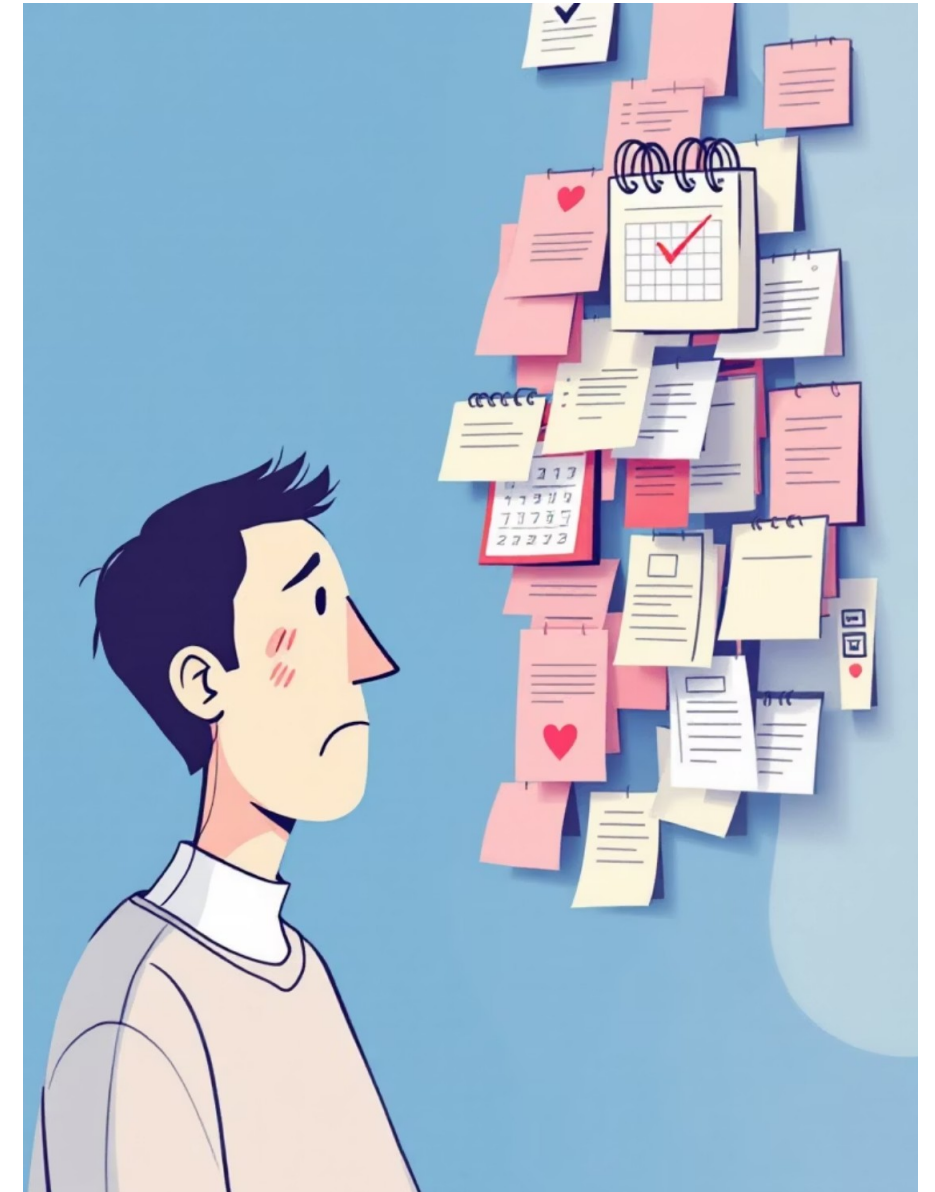
Streamlining court reservations through real-time booking



The Booking Challenge

Manual Process Pain Points

- Tracking bookings across multiple courts becomes increasingly difficult as demand grows
- Double bookings and schedule conflicts frustrate both customers and staff
- Admin management requires excessive time spent on phone calls and spreadsheets
- No real-time visibility into court availability leads to lost revenue opportunities

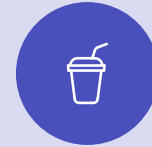


Project Vision



Seamless Booking

Users can browse available courts and reserve time slots instantly



Real-Time Updates

Live availability tracking prevents conflicts and ensures accuracy



Secure Platform

Protected data handling with authentication and secure transactions



Admin Control Panel

Centralized management for courts, schedules, and bookings

Technology Stack

Node.js & Express.js

- API routing and async server logic

MySQL Database

- Reliable relational data storage

React.js Frontend

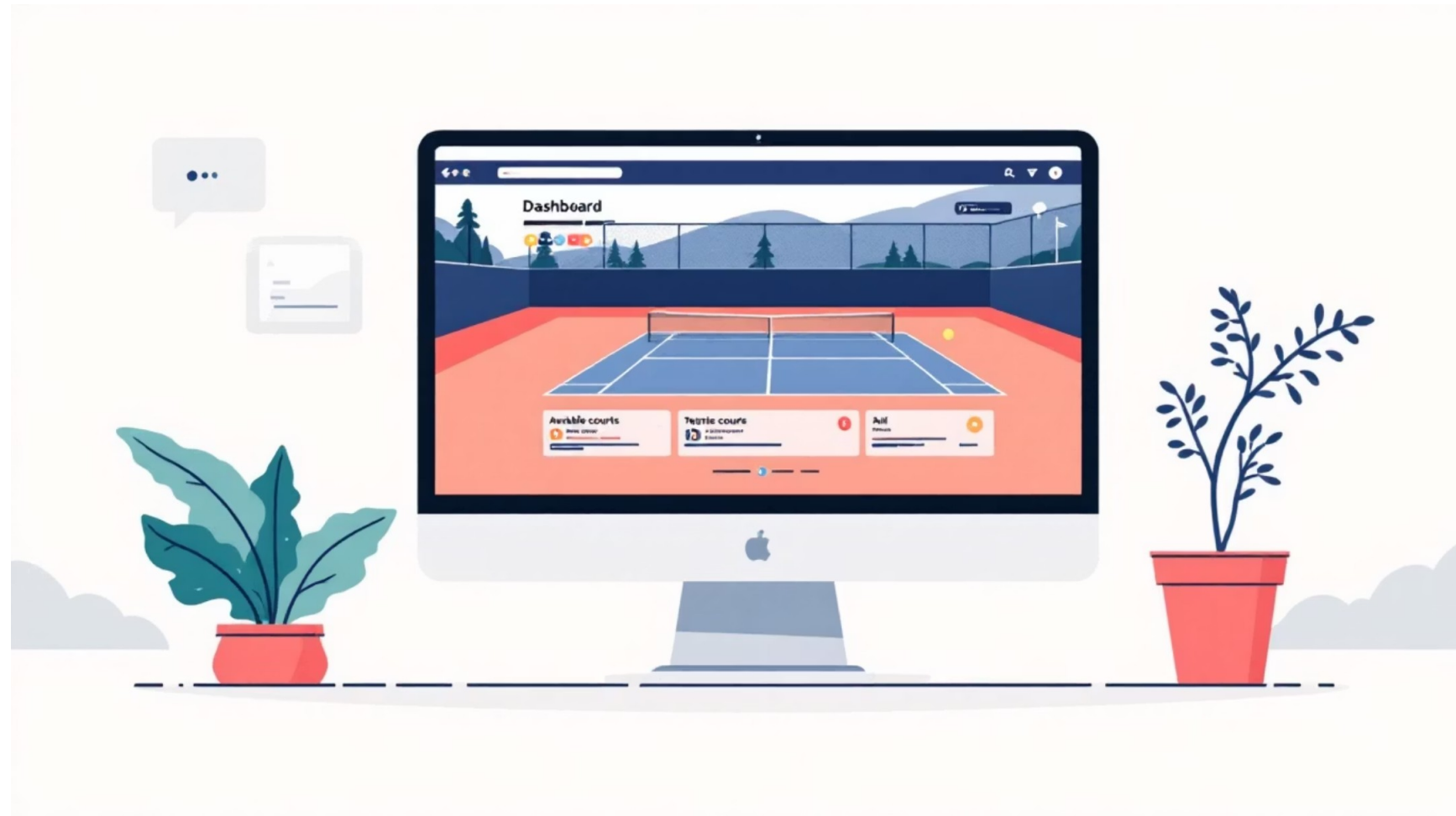
- Interactive component-based UI

CSS & Styling

- Consistent responsive visual design



System Modules: User Panel



User Module

- **Search & Filter Courts**
Find courts by location, price, amenities
- **Real-time Availability**
See live schedules and open slots
- **Easy Bookings & History**
Instant confirmation and reservation history

System Modules: Admin Panel

⚙️ Admin Module

- **Court Inventory Controls**

Manage capacity, pricing, and listings

- **Schedule Management**

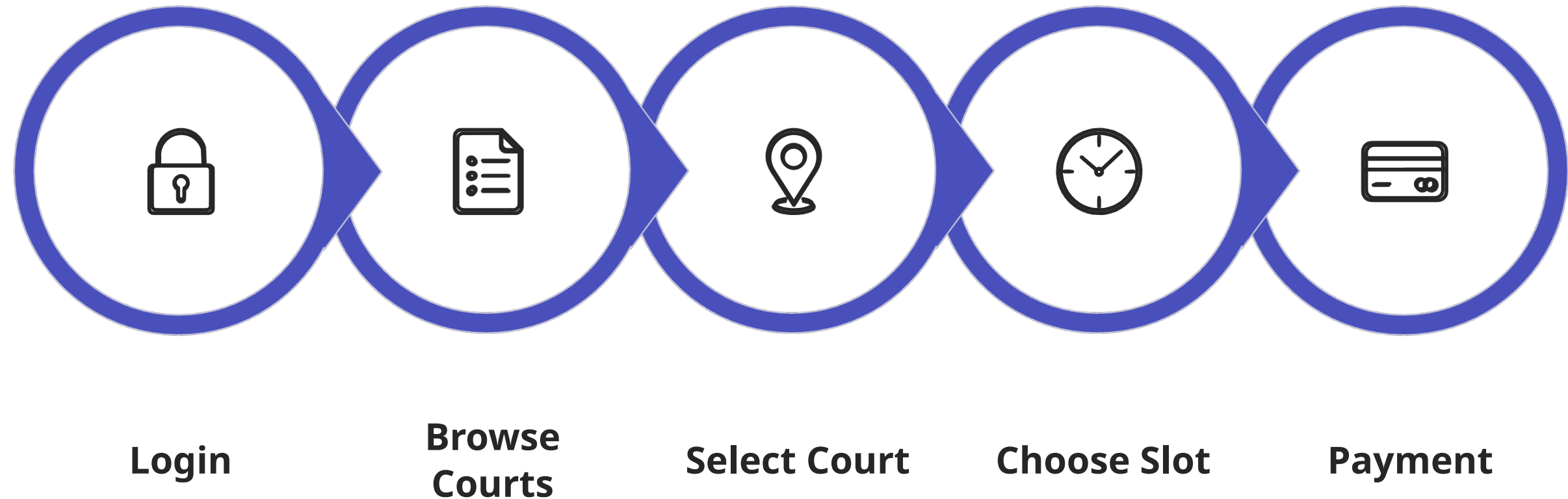
Bulk edits and timetable configuration

- **Analytics & Reporting**

Track bookings and platform metrics



System Workflow



Customer Journey

- Secure login to access personal account and bookings
- Browse courts with filters for location and amenities
- Select court and preferred time slot quickly
- Pay securely and receive instant booking confirmation

Admin Operations

- Log into admin panel with elevated access
- Configure court details, pricing, and availability
- Manage bookings and schedules via simple controls
- Monitor metrics and export reports for analysis

Future Enhancements



Payment Integration

Stripe or PayPal integration for seamless online transactions



Real-Time Notifications

WebSocket-based alerts for booking confirmations and reminders



Advanced Analytics

Dashboard with booking trends, revenue reports, and court utilization metrics

A Reliable, Efficient, and Scalable System for Modern Futsal Bookings

